

- Q.26 Describe the construction and working principle of plate heat exchanger.
- Q.27 Define diffusion and provide an example of a food processing application where diffusion plays an important role.
- Q.28 Define the Reynolds number and explain its significance in fluid mechanics.
- Q.29 Briefly explain the principle behind the operation of a centrifugal pump.
- Q.30 Discuss some key factors to consider when selecting a pasteurizer for a dairy processing plant.
- Q.31 Briefly Explain the operations of an evaporator in concentrating liquid food products.
- Q.32 How does a heat exchanger work to transfer heat between two fluids? Provide an example.
- Q.33 Describe the working principle of a dryer used in food processing and how it removes moisture from food products.
- Q.34 Briefly explain how a boiler generates steam for various food processing applications.
- Q.35 Define psychrometry and the concept of relative humidity in psychrometry.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Describe the operating principles of common drying methods used in the food industry, such as freeze drying and spray drying.
- Q.37 Discuss the modes of heat transfer, including conduction, convection and radiation and their application in food processing.
- Q.38 Define thermal conductivity, Viscosity, Density and explain their importance in food processing. Define organic foods.

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4th Semester/ Food Tech. Subject : Principles of Food Engineering

Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is a derived unit of measurement?
a) Meter (m) b) Kilogram (kg)
c) Newton ((N) d) Watt (W)
- Q.2 Porosity of a material refers to :
a) Its ability to conduct heat
b) The measure of its color intensity
c) Presence of holes or voids within the material
d) The resistance of the material to flow
- Q.3 Which of the following is commonly used to measure pressure in a fluid system?
a) Thermocouple b) Manometer
c) Tachometer d) Barometer
- Q.4 Reynolds number is a dimensionless quantity used to predict:
a) Pressure drop in a pipe
b) Fluid viscosity
c) Fluid Flow characteristics
d) Fluid density
- Q.5 In a positive displacement pump, the discharge flow rate is primarily determined by :
a) Pump speed b) Fluid viscosity
c) Pump size d) Inlet pressure

- Q.6 Conduction is the primary mode of heat transfer in :
 a) Liquids b) Gases
 c) Solids d) Vacuum
- Q.7 Which of the following is NOT a mode of heat transfer?
 a) Conduction b) Convection
 c) Radiation d) Evaporation
- Q.8 When selecting a pasteurizer for a dairy processing plant, which factor is typically the most critical?
 a) Energy efficiency b) Processing capacity
 c) Size of the equipment d) Availability of spare parts
- Q.9 What is the primary purposes of an autoclave in food processing?
 a) Rapid cooling of food products
 b) Heating food products to high temperature for preservation
 c) Mixing ingredients for food preparation
 d) Grinding food into fine particles
- Q.10 Which of the following is a common type of heat exchanger used in the food industry?
 a) Hydraulic pump b) Plate heat exchanger
 c) Conveyor belt d) Mixing tank

SECTION-B

- Note:** Objective/ Completion type questions. All questions are compulsory. (10x1=10)
- Q.11 Density is a physical property that can be measured in units such as grams per cubic centimeter. (True/False)
- Q.12 Specific gravity is a dimensionless quantity that compares the density of a substance to the density of a reference substance, typically water. (True/False)
- Q.13 Specific heat is the amount of heat required to raise the temperature of a unit mass of a substance of one degree Celsius or Kelvin. (True/False)

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- Q.14 Thermal diffusivity is a measure of how quickly heat spreads through a material relative to its ability to store heat. (True/False)
- Q.15 _____ are used to remove moisture from food products. (True/False)
- Q.16 _____ are used to sterilize food products by subjecting them to high pressure and temperature. (Pasteurizers / Autoclaves)
- Q.17 _____ are use to heat or cool food products by exchanging heat between two fluids. (Heat exchanges / Autoclaves)
- Q.18 _____ are use to raise the temperature of food products to a specific level for a certain to eliminate harmful bacteria. (Pasteurizers / Autoclaves)
- Q.19 _____ are crucial for heating or cooling food products during various stages of processing to maintain quality and safety. (Heat exchangers / Boilers)
- Q.20 The efficiency of heat transfer in a plate heat exchanger depends on the _____ of the plates. (Surface area / Radiation)

SECTION-C

- Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)
- Q.21 Why understanding of rheological properties of good materials in food processing is important?
- Q.22 Describe the principle of operation of a U-Tube manometer and explain how it is used to measure pressure in a fluid system.
- Q.23 Define convection and provide an example of heat transfer through conduction in a good process.
- Q.24 Describe how radiation heat transfer occurs and provide an example of a food processing application where radiation is utilized.
- Q.25 Name two types of heat exchangers commonly used in the food industry and briefly explain the operating principles of any one of them.

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